Khadijah Aamir Chaudhry

Computer Science · Secure Systems · Applied Research
Lahore, Pakistan · +92 345 4430787 · khadijah.aamir.ch@gmail.com
LinkedIn · GitHub · Portfolio

EDUCATION

Lahore University of Management Sciences (LUMS) *2023–2027*
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School *2018–2023*
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS *Aug 2025–present*
Supervisor: Dr. Muhammad Ali Siddiqi

- **Built a simulated Rocket Chip and Verilator testbed** to capture RISC-V transaction traces for hardware Trojan monitoring research.
- **Benchmarked three post-quantum algorithms** on ESP32 for latency, memory, network cost, and estimated energy.

Research Assistant, SysNet Lab, LUMS *Jan–May 2026*
Supervisor: Dr. Naveed Anwar Bhatti

- **Co-developed EyeSense's smartphone ToF and machine-learning pipeline** to explore accessible eyesight screening.
- **Supported lab and eye-camp data collection** for a 120-participant team study that reported 1.15 D mean absolute error.

SELECTED PROJECTS

On Chip Behavioral Monitoring · write-up *2026–present*

- **Set up four RV32 configurations** and captured transaction traces with Verilator for runtime monitoring research.
- **Replayed benign and trigger-active firmware** to create traces for later detector evaluation.

EyeSense · write-up *Jan–May 2026*

- **Co-developed a ToF phase-map and machine-learning pipeline** to explore phone-based refractive-error screening.
- **Contributed to a 120-participant team evaluation** that reported 1.15 D mean absolute error for eyesight screening.

Post-Quantum Cryptography on ESP32 · code *Aug–Dec 2025*

- **Benchmarked Kyber-512, Light-SABER, and Dilithium-II** on ESP32 to assess cryptography on low-resource hardware.
- **Compared latency, memory, network cost, and estimated energy** to clarify trade-offs for embedded use.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS *Aug 2026–present*

- **Designed and conducted programming labs** and prepared and graded quizzes for about 450 first-year students.
- **Held weekly office hours and invigilated exams** to provide recurring course support.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions *Jul 2026–present*

- **Coordinated team delivery of the brand kit and office collateral** for the consultancy.
- **Coordinated LinkedIn launch content and the website** so prospective clients can understand services and get in touch.

Data Collection Coach, Uplift AI *Jul–Aug 2024*

- **Led 10 data collectors and organized field outreach** for a Pakistani-language AI dataset.
- **Ran daily quality reviews and feedback sessions** to improve consistency in collected material.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS *Aug 2026–present*

- **Led orientation and faculty introductions** to help 15 incoming students adjust to university.
- **Continued monthly first-year support sessions** to offer students an ongoing point of contact.

Women in STEM Coordinator, LUMS SPADES *Sep 2024–May 2025*

- **Helped launch SPADES' first women-only STEM event** at PSIFI to widen participation in STEM.
- **Engaged more than 100 participants** in a dedicated STEM program.

Co-Founder and Logistics Director, Umeed-e-Nau NGO *Sep 2021–May 2023*

- **Co-founded an NGO** to create inclusive learning opportunities for underprivileged and disabled children.
- **Organized educational activities and visits** that enabled children to learn and participate together.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

100% Merit Scholarship, Lahore Grammar School · *Aug 2021–May 2023*
Shortlisted writing submission, LUMSx Nexus · *May 2025*
Honorable Mention, EducationUSA Pakistan Writing Contest · *Sep 2022*

TECHNICAL SKILLS

Programming: C · C++ · Python · JavaScript · Java · Assembly
Embedded and systems: ESP32 · Tiva C · MIPS · RISC-V · Rocket Chip · Verilator · FreeRTOS
Web and mobile: React · Android · Firebase Authentication · Firestore
Methods and tools: ML/AI · CNNs · PCA · K-means · ToF sensing · Docker · Git · post-quantum benchmarking